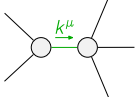
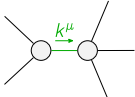


	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{K}_{0^+}^{\#1} \dagger$	0	0				
$\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$-k^2 \frac{(4)}{\kappa_3}$	$\frac{k^2 \frac{(4)}{\kappa_4}}{2 \sqrt{2}}$	0	0		
$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{k^2 \frac{(4)}{\kappa_4}}{2 \sqrt{2}}$	$\frac{\Upsilon_1 k^2}{2}$	0	0		
$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	$k^2 \frac{(4)}{\kappa_1}$	$\sqrt{2} k^2 \frac{(4)}{\kappa_1}$		
$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	$\sqrt{2} k^2 \frac{(4)}{\kappa_1}$	$2 k^2 \frac{(4)}{\kappa_1}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
	$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$				0	0
	$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$				0	0

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{T}_{0^+}^{\#1} \dagger$	0	0				
$\mathcal{T}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{\Upsilon_1 k^2}{2 \text{Det}(1^+)}$	$-\frac{k^2 \binom{4}{\kappa_4}}{2 \sqrt{2} \text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{k^2 \binom{4}{\kappa_4}}{2 \sqrt{2} \text{Det}(1^+)}$	$-\frac{k^2 \binom{4}{\kappa_3}}{\text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	$\frac{1}{9 k^2 \binom{4}{\kappa_1}}$	$\frac{\sqrt{2}}{9 k^2 \binom{4}{\kappa_1}}$		
$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	$\frac{\sqrt{2}}{9 k^2 \binom{4}{\kappa_1}}$	$\frac{2}{9 k^2 \binom{4}{\kappa_1}}$	$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
			$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0	
			$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	

Abbreviations used in matrices			
$\Upsilon_1 == 9 \frac{(4)}{\kappa_1} + \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4}$ && $\text{Det}(1^+) == -\frac{1}{8} k^4 (36 \frac{(4)}{\kappa_1} \frac{(4)}{\kappa_3} + (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)$			
Lagrangian			
$\kappa_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta - \kappa_1 \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \kappa_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi}{}^\beta + \kappa_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi}{}^\alpha - \kappa_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha}{}^\beta +$ $6 \frac{(4)}{\kappa_1} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta \partial_\delta \mathcal{K}^\delta_{\chi\beta}{}^\alpha + \frac{9}{2} \frac{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi}{}^\alpha + \frac{1}{2} \frac{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi}{}^\alpha + \frac{1}{2} \frac{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi}{}^\alpha$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}^\alpha{}^\beta{}_\alpha == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} == \mathcal{J}_{1^+}^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta == 0$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^\delta{}_\delta{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha}{}_\delta ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^\delta{}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta}{}_\delta$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^\chi{}_\chi{}^\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^\chi{}_\chi{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	15		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{48 \frac{(4)}{\kappa_1}{}^2 + 8 \frac{(4)}{\kappa_1} (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4}) + (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2}{\frac{(4)}{\kappa_1} (36 \frac{(4)}{\kappa_1} \frac{(4)}{\kappa_3} + (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)}$
	1	0	$-\frac{216 \frac{(4)}{\kappa_1}{}^2 + 24 \frac{(4)}{\kappa_1} (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4}) + (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2}{\frac{(4)}{\kappa_1} (36 \frac{(4)}{\kappa_1} \frac{(4)}{\kappa_3} + (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)}$

Resolved unitarity condition(s):

$$(\frac{(4)}{\kappa_4} < 0 \&\& ((\frac{(4)}{\kappa_3} < 0 \&\& (\frac{(4)}{\kappa_1} < 0 \parallel \frac{(4)}{\kappa_1} > \frac{-4 \frac{(4)}{\kappa_3}{}^2 - 4 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_4} - \frac{(4)}{\kappa_4}{}^2}{36 \frac{(4)}{\kappa_3}}))) \parallel (\frac{(4)}{\kappa_3} == 0 \&\& \frac{(4)}{\kappa_1} < 0) \parallel$$

$$(0 < \frac{(4)}{\kappa_3} < -\frac{\frac{(4)}{\kappa_4}}{2} \&\& \frac{-4 \frac{(4)}{\kappa_3}{}^2 - 4 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_4} - \frac{(4)}{\kappa_4}{}^2}{36 \frac{(4)}{\kappa_3}} < \frac{(4)}{\kappa_1} < 0) \parallel ((\frac{(4)}{\kappa_3} > -\frac{\frac{(4)}{\kappa_4}}{2} \&\& \frac{-4 \frac{(4)}{\kappa_3}{}^2 - 4 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_4} - \frac{(4)}{\kappa_4}{}^2}{36 \frac{(4)}{\kappa_3}} < \frac{(4)}{\kappa_1} < 0))) \parallel$$

$$(\frac{(4)}{\kappa_4} == 0 \&\& ((\frac{(4)}{\kappa_3} < 0 \&\& (\frac{(4)}{\kappa_1} < 0 \parallel \frac{(4)}{\kappa_1} > -\frac{\frac{(4)}{\kappa_3}}{9}))) \parallel (\frac{(4)}{\kappa_3} > 0 \&\& -\frac{\frac{(4)}{\kappa_3}}{9} < \frac{(4)}{\kappa_1} < 0))) \parallel$$

$$(\frac{(4)}{\kappa_4} > 0 \&\& ((\frac{(4)}{\kappa_3} < -\frac{\frac{(4)}{\kappa_4}}{2} \&\& (\frac{(4)}{\kappa_1} < 0 \parallel \frac{(4)}{\kappa_1} > \frac{-4 \frac{(4)}{\kappa_3}{}^2 - 4 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_4} - \frac{(4)}{\kappa_4}{}^2}{36 \frac{(4)}{\kappa_3}}))) \parallel$$

$$(\frac{(4)}{\kappa_3} == -\frac{\frac{(4)}{\kappa_4}}{2} \&\& (\frac{(4)}{\kappa_1} < 0 \parallel \frac{(4)}{\kappa_1} > 0)) \parallel (-\frac{\frac{(4)}{\kappa_4}}{2} < \frac{(4)}{\kappa_3} < 0 \&\& (\frac{(4)}{\kappa_1} < 0 \parallel \frac{(4)}{\kappa_1} > \frac{-4 \frac{(4)}{\kappa_3}{}^2 - 4 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_4} - \frac{(4)}{\kappa_4}{}^2}{36 \frac{(4)}{\kappa_3}}))) \parallel$$

$$(\frac{(4)}{\kappa_3} == 0 \&\& \frac{(4)}{\kappa_1} < 0) \parallel ((\frac{(4)}{\kappa_3} > 0 \&\& \frac{-4 \frac{(4)}{\kappa_3}{}^2 - 4 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_4} - \frac{(4)}{\kappa_4}{}^2}{36 \frac{(4)}{\kappa_3}} < \frac{(4)}{\kappa_1} < 0)))$$